METHODOLOGY | LATEST COMPLETED SIGNAL JULY 31, 2026

Macro Seasons v4

How point-in-time macro evidence becomes monthly season probabilities and two investable ETF portfolios.

The model reads growth, inflation and liquidity at each completed month-end. Growth and inflation define four economic seasons. Liquidity changes how much risk the portfolio takes. The output is a probability mix across all four seasons, so one noisy release changes the portfolio gradually.

1 The four seasons

Season	Growth and inflation	Economic reading	Fixed sleeve emphasis
SPRING	Growth rising Inflation cooling	Recovery with disinflation	Broad equities, technology, credit, housing and modest duration
SUMMER	Growth rising Inflation heating	Reflationary expansion	Cyclicals, value, energy, materials, emerging markets, commodities and TIPS
FALL	Growth falling Inflation heating	Stagflation	Gold, broad commodities, inflation protection, dollar, defensive equities and short duration
WINTER	Growth falling Inflation cooling	Disinflationary slowdown	Treasuries, defensive and low-volatility equities, gold, safe-haven currencies and cash

2 What enters the signal

GROWTH

Is activity gaining or losing momentum?

Industrial production and payrolls use ALFRED vintages. Jobless claims use an explicit release lag. Cyclical versus defensive equities and equities versus intermediate Treasuries confirm the reading.

Main windows: 3 to 6 months

INFLATION

Is inflation pressure building or cooling?

CPI momentum uses ALFRED vintages. Five-year breakevens and oil use release-lagged FRED data. TIPS relative to nominal Treasuries adds market confirmation.

Main windows: 3 to 12 months

LIQUIDITY

Are financial conditions adding or removing support?

M2 uses ALFRED vintages. Fed net liquidity, financial conditions, high-yield spreads and the two-year yield use release-lagged FRED data to capture money, credit and policy conditions.

Role: changes risk intensity

3 The monthly decision path

1. Cut the data
Freeze the information available at the completed month-end.

2. Build the pillars
Measure trailing change, standardize it against prior history and combine the sub-signals.

3. Set probabilities
Growth and inflation distribute weight across all four seasons. No hard regime switch is required.

4. Adjust risk
Liquidity, real rates, credit, momentum and trend alter the blended portfolio within fixed limits.

5. Trade next month
Month-end weights govern the following calendar month's return. Turnover costs are charged.

6. Lock the record
Later data and revisions cannot change an earlier decision in the prefix-causality tests.

Causal has a narrow meaning in this research. A decision for the next month can depend only on information available by the current month-end. Appending future observations must leave every earlier signal, weight and return unchanged. The model does not claim that a macro pillar proves economic causation.

CPI, industrial production, payrolls and M2 use ALFRED vintages. Other FRED inputs use explicit publication lags with latest-revised history. Yahoo adjusted prices supply the traded series. The latest production refresh passed 49/49 freshness checks.

How to read a season percentage
The percentages are soft classification weights. A 31% Summer reading means the current growth and inflation scores assign 31% of the season blend to Summer. It is not a calibrated 31% chance that a future event will occur.

Time horizon and stability
The reading describes the macro state at the completed month-end and sets the next calendar month's portfolio. It does not predict how long the season will last. A close top-two split keeps both sleeves active instead of forcing a full switch.

Symmetric scoring
Each sub-signal is compared with its own trailing 10-year history after a minimum 36-month warm-up. Extreme readings are capped and the pillar average receives light smoothing. Growth and inflation are then mapped symmetrically, so no season receives a built-in advantage.

PORTFOLIO CONSTRUCTION AND WALK-FORWARD EVIDENCE

Two portfolios, one macro engine

Both tracks start from the same monthly season probabilities. Their risk budgets, return sources and implementation costs are different.

LONG-ONLY SEASON PORTFOLIO

A fully funded ETF allocation designed to reduce drawdowns while retaining exposure to the active macro mix.

1. Blend the four fixed season sleeves using the current probabilities.
2. Combine each sleeve's economic weights with a trailing 36-month inverse-volatility tilt.
3. Let liquidity shift a bounded share between risky and defensive assets.
4. Rotate the defensive sleeve between gold and nominal duration as real yields change.
5. Halve risky positions when credit spreads are both high and widening.
6. Apply relative momentum measured from 12 months ago to one month ago, followed by the 200-day trend gate. Failed trend positions move to BIL.
7. Target 10% volatility from trailing 24-month risk. Scaling can reduce exposure; this track does not borrow.

Backtest cost: 10 basis points per unit of monthly turnover.

L/S PORTFOLIO

A separate 10% volatility strategy combining macro allocation with an independent trend-following return stream.

1. Start with a risk-scaled core season stream and a risk-scaled enhanced long-only stream.
2. Add time-series momentum across 13 liquid ETFs covering equities, rates, credit, real assets and currencies.
3. Hold an ETF long when its trailing 12-month return beats BIL and short when it trails BIL.
4. Size trend positions from trailing 36-month volatility and cap each position at 20% of the sleeve.
5. Weight the three streams from their trailing 24-month volatility, shifted one month, then target 10% total volatility.
6. Aggregate duplicate ETF exposures, net longs against shorts, use BIL before margin borrowing and enforce gross, net, short and ticker limits.

Cost model: commissions, regulatory fees, 1 bp slippage, financing, stock borrow and interest on short proceeds.

4 Portfolio design and walk-forward discipline

How the ETF sleeves were chosen
The season templates were set from economic exposure, liquidity and trading history. They were not selected by maximizing returns in the same season samples. Equity beta and cyclicals express growth; Treasuries and defensive equities express slowdown; TIPS, commodities and gold express inflation; BIL and major currencies provide defense or funding. Pre-inception mutual funds and futures extend selected ETF histories and introduce basis risk.

What walk-forward means
At each month-end the model rebuilds the decision from that date's data, sets weights, and applies them to the next month's returns. Rolling risk estimates and stream weights use prior months and are shifted before use. Turnover and financing costs are charged as positions change. Prefix tests rerun the model at earlier cutoffs and require the old path to match exactly.

5 Historical evidence after the stated cost rules

Historical simulation	Window	CAGR	Volatility	Excess Sharpe	Max drawdown
Long-only season portfolio	Jan 2000 - Jul 2026	5.57%	4.53%	0.79	-5.48%
L/S portfolio after IBKR costs	Jan 2007 - Jul 2026	7.79%	8.12%	0.75	-11.93%
S&P 500	Jan 2000 - Jul 2026	8.20%	15.23%	0.40	-50.78%
60/40 SPY/AGG	Jan 2000 - Jul 2026	6.70%	9.52%	0.49	-32.32%

The website's 9.21% return-level ensemble represents the target streams before physical netting and explicit IBKR financing. The holdings-based, costed L/S record begins in January 2007 and earns 7.79% over the frozen sample.

6 Risk and operating controls

Long-only boundary
Weights remain fully funded. The 10% volatility target acts as a ceiling because exposure can be reduced but cannot exceed 100%. Trend failures move to BIL. This track has no short positions or margin debit.

L/S hard limits
Gross exposure is capped at 175%, net exposure at 150%, gross shorts at 35%, each non-cash ETF at 25%, and margin debit at 50%. The whole book still requires an IBKR Portfolio Margin preview.

Monthly operation
The completed month-end signal is frozen before the next allocation. Current-month prices update performance only and cannot alter that signal. Stale data, missing vintages or failed causality tests stop deployment.

What the evidence supports
The historical record tests a fixed decision process across repeated monthly vintages. It supports comparison of drawdown, volatility and return after the stated cost rules. The 2019-2026 period has already informed research review and is not presented as an untouched holdout.

Material limits
Most non-core FRED series use publication lags with revised history. Yahoo prices are a public research feed. Proxy splices can differ from later ETFs. Borrow availability, taxes and market impact are modeled imperfectly. Portfolio Margin must be checked on the live IBKR account before trading.

Sources: Federal Reserve FRED and ALFRED, Yahoo Finance adjusted prices, and published IBKR Pro commission and financing schedules. Full source, immutable V4 inputs, hashes and monthly artifacts are retained in the release package. Historical simulations are research results, not investment advice.